

PROJECT_ANALYSIS.md — KrishiMitra (कृषिमित्र) AI Agriculture Assistant

1. Executive Summary & Environment Assessment

An exhaustive inspection of the host system and workspace was conducted prior to code implementation.

System Diagnostics

- Operating System: Windows 11 (AMD64)
- CPU: Intel Core Ultra 7 155H
- GPU: NVIDIA GeForce RTX 4060 Laptop GPU (8GB GDDR6 VRAM, CUDA Driver 13.3, SMI 610.62)
- System RAM: 16 GB
- Java Development Kit: Microsoft OpenJDK 21.0.12.8 LTS (**C:\Program Files\Microsoft\jdk-21.0.12.8-hotspot\bin\javac.exe**)
- Android SDK: Present at **C:\Users\vibho\AppData\Local\Android\Sdk**
- Platforms available: **android-37.0** (API 35/36/37 compatible)
- Build Tools: **36.0.0**
- Platform Tools: **adb.exe** operational
- Emulator & AVD: **Pixel_8** AVD configured
- Android Studio: Installed with plugins and JetBrains Runtime at **C:\Program Files\Android\Android Studio**
- Gradle: Gradle 9.5.0 binary and cache present at **C:\Users\vibho\gradle\wrapper\dists**
- Python Environment:
- Python 3.13.14 (Active in PATH) & Python 3.14
- Pre-installed packages: **fastapi, uvicorn, pydantic, sqlalchemy, httpx, pillow, pytest, python-dotenv**
- Machine learning packages ready for installation via standard wheels: **numpy, scikit-learn, scipy, onnx, onnxruntime, torch, torchvision**
- Existing Project Files: Workspace was initialized clean (**c:\Users\vibho\OneDrive\Desktop\Farmer Android App**).

2. Product Identity & Branding

- Selected Brand Name: KrishiMitra (कृषिमित्र)
- Tagline: "आपका डिजिटल कृषि साथी — Your Intelligent Farming Companion"
- Target Audience: Indian smallholder and marginal farmers, agricultural extension workers, and rural youth operating budget Android smartphones (2GB–4GB RAM, weak 4G/2G, intermittent connectivity).
- Design Philosophy: High-contrast, touch-friendly UI, Hindi-first bilingual architecture, zero unnecessary visual bloat, instant offline responsiveness (< 25ms NLP, < 75ms vision inference).

3. Detected Requirements vs. Architecture Matrix

Requirement	Proposed Architecture Component	Offline / Online Strategy
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Agricultural Q&A	Room DB + On-device NLP Intent Engine + TF-IDF Vectorizer	100% Offline Primary; FastAPI Cloud AI Fallback when online & low confidence
Crop Disease Vision	CameraX + Quantized MobileNetV2/V3 TFLite Model (12 Major Indian Crop Diseases)	100% Offline (On-device neural inference)
Voice Assistant	Hold-to-Talk + Android Native SpeechRecognizer + TTS Engine	Hybrid: Native offline STT/TTS with backend Whisper/TTS API fallback
Government Schemes	Room DB + Backend Sync API (/schemes)	Offline Cached with verified provenance (PM-KISAN, PMFBY, KCC, etc.)
Agri Loans	Room DB + Backend Sync API (/loans)	Offline Cached with verified criteria (KCC, NABARD, Gold Loans)
Weather & Agri Advisory	Open-Meteo / IMD Weather Service + Local Rule Advisory Engine	Cached offline with GPS & manual district selector
Synchronization	Differential REST Sync Engine (/sync)	Auto-detects connectivity; syncs latest schemes, crops, and advisory

4. Key Subsystem Design

Subsystem 1: Android Application (android/)

- Language & UI: Kotlin 2.1+, Jetpack Compose, Material 3, Accompanist, Navigation Compose.
- Architecture: Clean Architecture + MVVM (Model-View-ViewModel) with StateFlow.
- Local Persistence: Room SQLite Database preloaded with verified agricultural facts, crops, schemes, and loans from assets.
- Hardware Interop: CameraX with custom Leaf Viewfinder, Android SpeechRecognizer, TextToSpeech, FusedLocationProviderClient with manual district fallback.
- Localization: Complete string externalization in English (values/strings.xml) and Hindi (values-hi/strings.xml).

Subsystem 2: Offline ML & NLP Pipeline (ml_pipeline/)

- Intent & Retrieval Model: Dual-stage lightweight pipeline:
 1. Agriculture domain intent & crop entity classifier (Cereals, Pulses, Soil, Irrigation, Pests, Schemes, Loans).
 2. N-gram TF-IDF cosine similarity search over local Room knowledge database.
 3. Latency: < 15ms on low-end ARM Cortex-A53 / A55 CPUs.
- Crop Disease Vision Model:
 1. 12 Indian agricultural classes: Rice Blast, Rice Brown Spot, Wheat Rust, Wheat Loose Smut, Cotton Bacterial Blight, Potato Early Blight, Potato Late Blight, Tomato Early Blight, Tomato Leaf Mold, Healthy Leaf, Soil/Background, Low Quality/Uncertain.
 2. Quantized to INT8/FP16 TFLite (< 4MB file size).
 3. Preprocessing: 224x224 normalization, fast TensorBuffer inference.

Subsystem 3: Backend API Server (backend/)

- Framework: FastAPI (Python 3.13) with asynchronous endpoints.
- Database: SQLite / PostgreSQL with SQLAlchemy 2.0 ORM.
- Cloud AI Fallback: AIProvider interface with LocalFallbackProvider, OpenAIGeminiCompatibleProvider, and GeminiProvider support.

- Sync & Seed: Endpoints for full data updates, schema versions, and knowledge attribution.
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5. Phased Implementation Sequence

1. Phase 1: Environment & Project Analysis (Completed)
2. Phase 2: Android Project Foundation & Gradle Configuration
3. Phase 3: Localization Architecture (Hindi & English)
4. Phase 4: Verified Agricultural Datasets & Generation Pipeline
5. Phase 5: Offline Room Database & Seed Loading
6. Phase 6: Lightweight NLP Intent & Retrieval Engine
7. Phase 7: Crop Disease Computer Vision Model Training & Quantization
8. Phase 8: CameraX & On-Device Vision Integration
9. Phase 9: Voice Assistant Pipeline (Hold-to-Talk)
10. Phase 10: Weather & Agri-Advisory Engine
11. Phase 11: Government Schemes & Loan Information Hub
12. Phase 12: Backend FastAPI Server & Cloud AI Fallback
13. Phase 13: Offline Data Synchronization
14. Phase 14: Comprehensive Verification, Testing & Debug APK Build
15. Phase 15: Documentation Suite & Delivery